

What is a brake booster?



A [Brake Booster](#) is a device that increases the force applied from the brake pedal to the master cylinder. The booster is what separates a power-assisted braking system from a manual one.

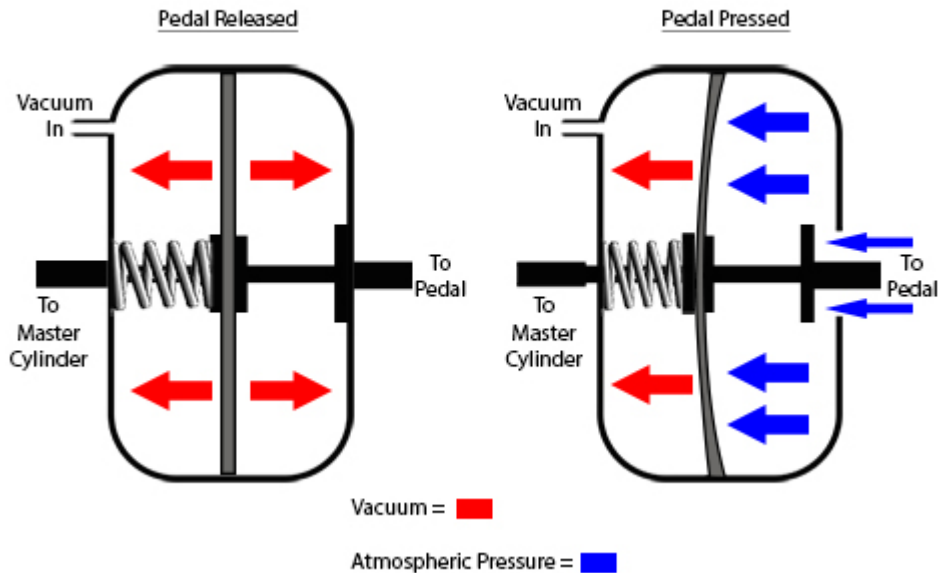
The most common type of brake booster uses [Engine Vacuum](#). However, hydraulic boosters are also available. They use fluid pressure created by the power steering pump. Some vehicles also use an electric powered brake booster.

How does it work?

A vacuum booster is usually mounted on the firewall, in the engine compartment. It has a chamber divided in half by a diaphragm. The chamber is connected to a vacuum source, usually the intake manifold. It also has a shaft with valves running through the center.

The brake pedal is linked to one side of the booster. The [Master Cylinder](#) is mounted on the other side.

Vacuum Brake Booster



Before you push the brake pedal, both sides of the diaphragm are supplied with vacuum.

When you push the pedal:

The shaft moves forward and opens the valve on the back of the chamber.

Atmospheric pressure enters half of the chamber.

The other half of the chamber is still under vacuum.

The difference in pressure moves the shaft forward even more.

The shaft then presses on the master cylinder pushrod.

When you release the pedal, a spring returns the shaft to its starting point. This also returns the valves to their starting points. Vacuum is then equalized on both sides of the diaphragm.

How does it affect performance?

The brake booster pushes harder on the master cylinder pushrod. In many cases, it can add 200–300 lbs. of force. However, the actual force depends on:

The size of the diaphragm

The amount of vacuum applied

The atmospheric pressure in the air

The added force helps create the high pressure needed in the braking system without increasing the force you apply to the pedal. This makes stopping easier and driving more enjoyable.

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